VLSI based Design and Implementation of Digital Baseband Processor for BLE

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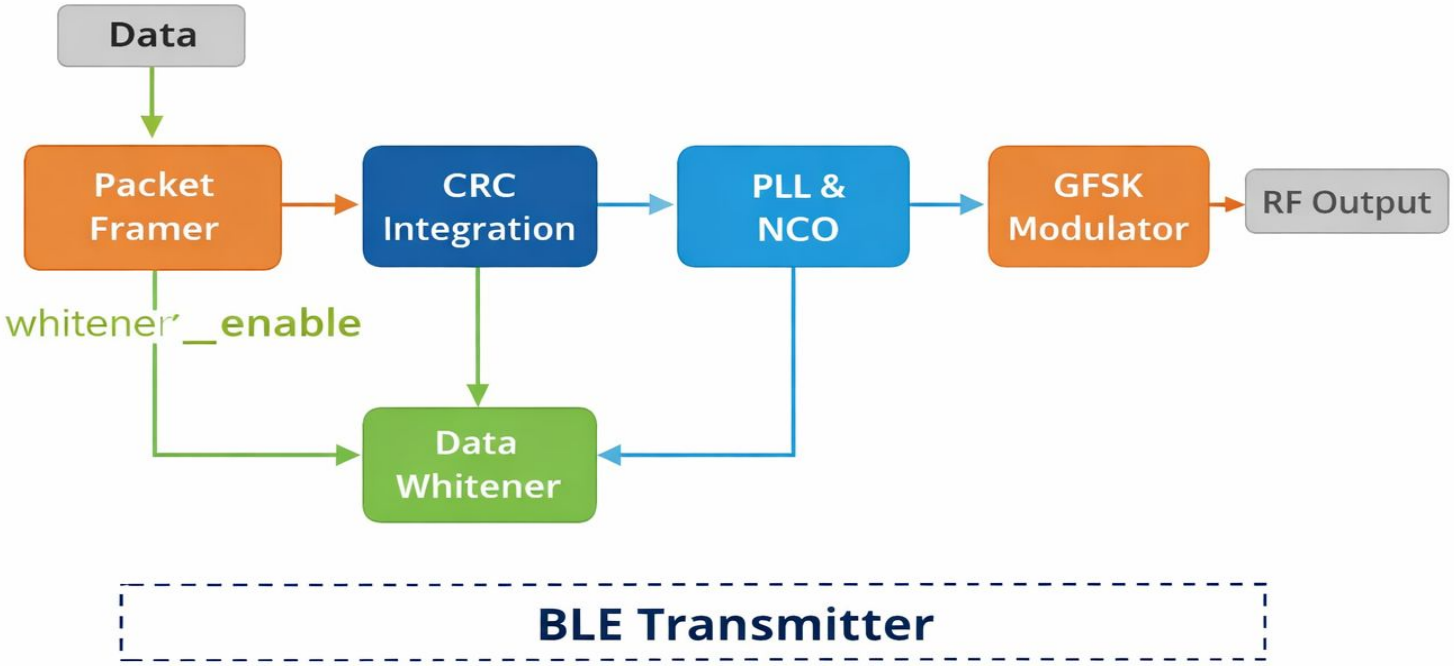
Introduction

- Bluetooth Low Energy (BLE) is a low-power wireless protocol used in IoT devices.
- The BLE baseband transmitter performs packet formation, CRC, whitening, and modulation.
- Efficient digital design helps reduce power and hardware complexity.
- So far, we have implemented the complete BLE TX chain: Packet Framer, CRC integration, Data Whitener, PLL, and GFSK Modulator.
- Future work includes receiver design, hardware implementation, and power optimization.

Introduction



Motivation

- The growth of IoT and wearable systems demands compact and energy-efficient wireless communication.
- BLE offers low power and reliable data exchange, making it ideal for portable devices.
- Efficient baseband processor design is essential to minimize power and area while maintaining performance.
- Existing BLE implementations are often not optimized for hardware-level efficiency.
- This work aims to develop a custom low-power BLE baseband processor optimized for future IoT applications.

Problem Statement with Objectives

Problem Statement-

The project aims to design a low-power BLE digital baseband processor for efficient packet processing, CRC generation, and data storage. It focuses on reducing power and area consumption while ensuring reliable data transfer for IoT communication systems.

Objectives-

- Integrate the previously designed CRC-24 generator into the BLE transmission chain.
- Design and implement a BLE-compliant Packet Framer with dynamic payload handling.
- Implement a 7-bit LFSR-based Data Whitener as per BLE specification.
- Develop a digital NCO-based GFSK Modulator with PLL-based clock scaling.
- Verify the complete transmitter chain through Verilog simulation and waveform analysis.

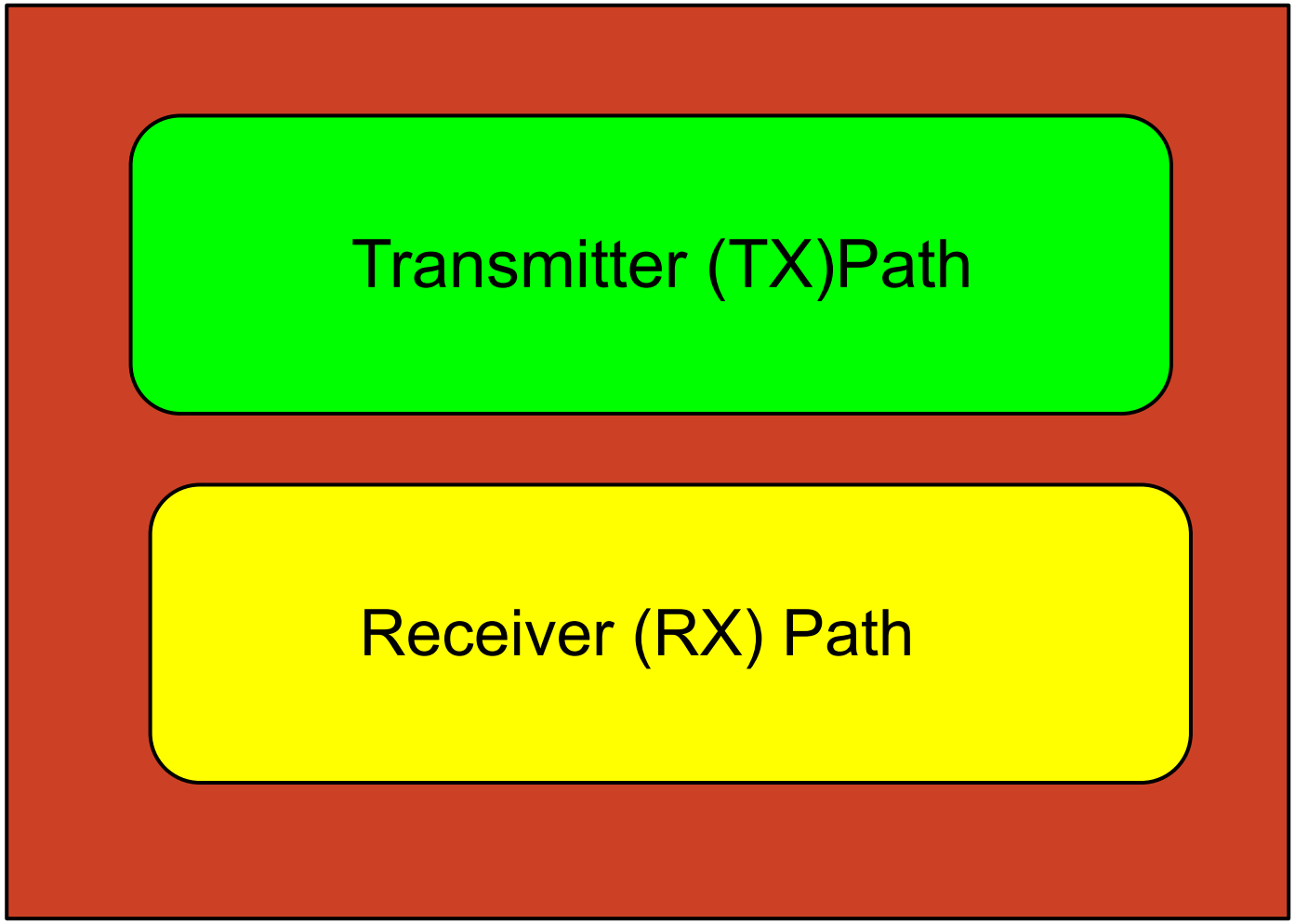
Literature Review

Sr no.	Title	Year Of Publication	Benefits	Limitations
[1]	Area-Efficient Digital Baseband Module for Bluetooth Wireless Communications (KAIST)	2004	Compact digital baseband module in 0.25 µm CMOS (≈2.25 mm²), synthesizable, supports Bluetooth baseband tasks in hardware.	Many baseband functions offloaded to microcontroller software; not fully digital; old CMOS process.
[2]	Bluetooth Technology: Principle, Applications and Current Status (Thapar Univ.)	2006	Provides ctooth protocols, profiles, good referenceomprehensive overview of Blue for Bluetooth fundamentals.	Survey rather than hardware implementation; lacks power/process metrics.

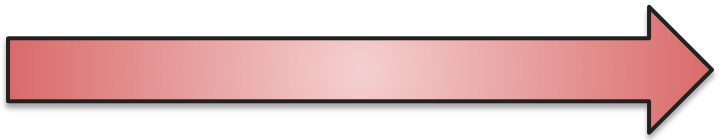
Literature Review

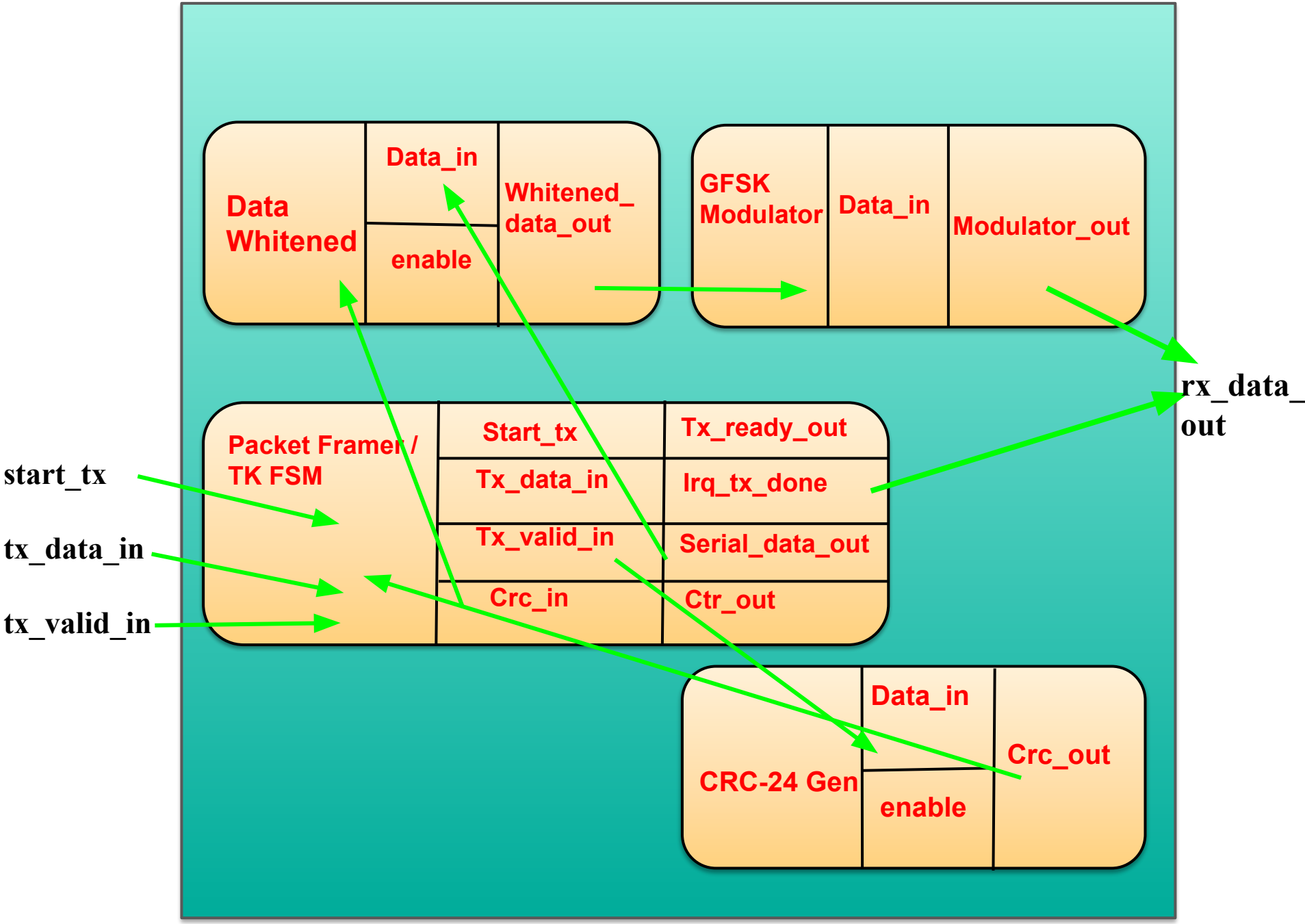
Sr no.	Title	Year Of Publication	Benefits	Limitations
[3]	A Baseband Processor for Impulse Ultra-Wideband Communications (MIT/Chandrakasan)	2005	Fast digital processing with integrated ADC and parallel architecture.	UWB-specific with low data rate, not suited for BLE/NB-IoT.
[4]	Low Power Bluetooth Single-Chip Design	2010	Ultra-low power Bluetooth chip optimized for mobile devices.	Covers only transmitter; lacks full baseband design.
[5]	A Low-Power Circuit Design of BLE Baseband Transmission Data Processing Module	2024	Implements CRC, whitening, and coding modules in Verilog; achieves low power and small area.	Limited to simulation results; no real hardware validation.

Block Diagram



Transmitter (TX) Path

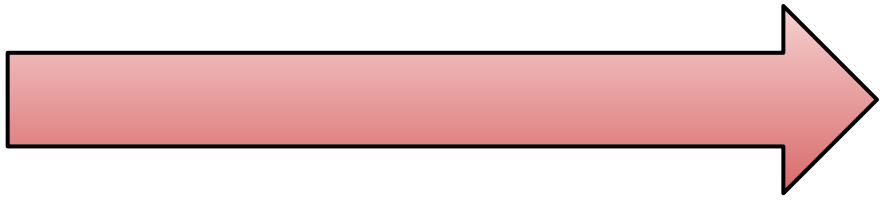


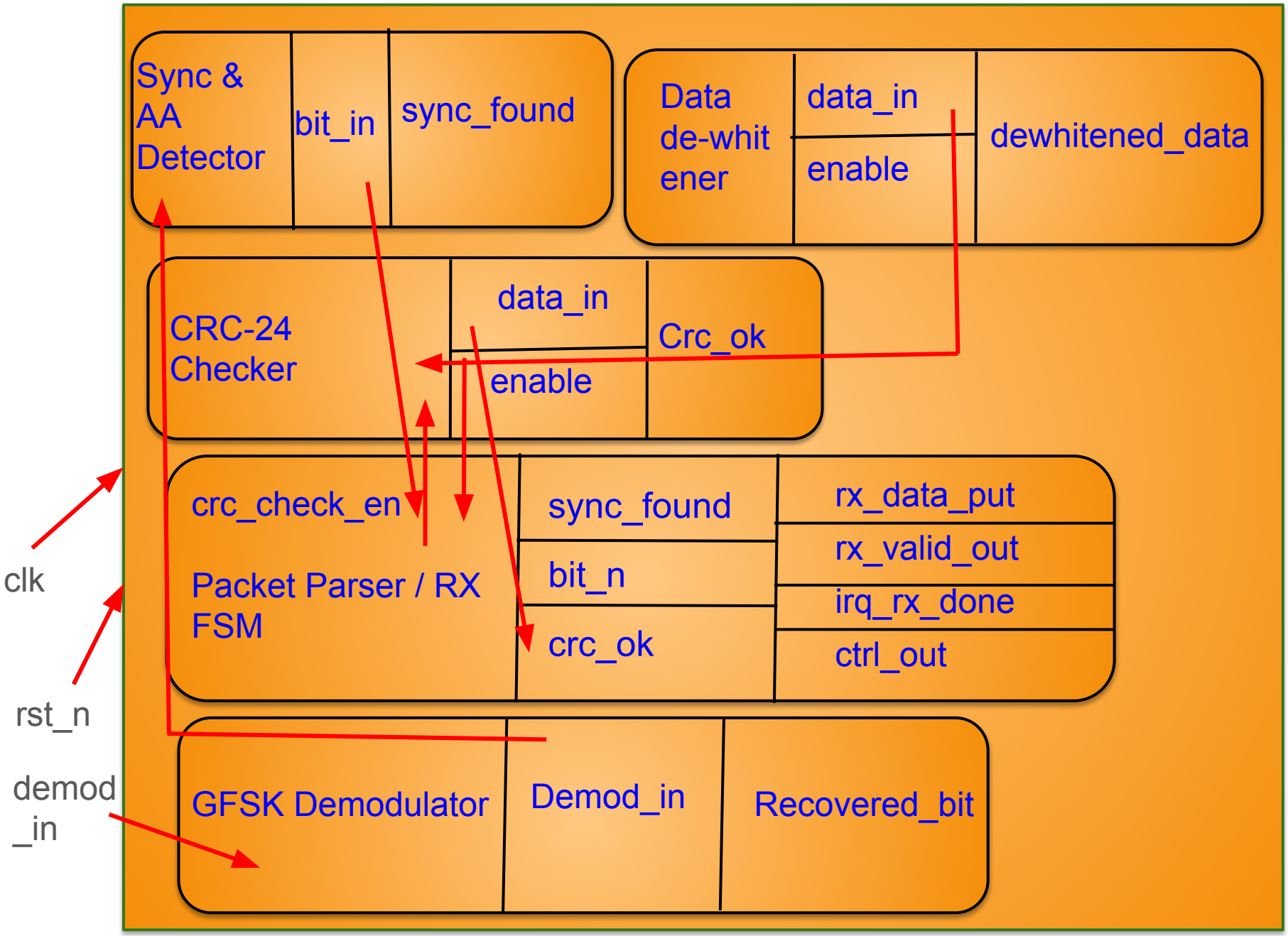

Data Whitened
Data_in
enable
Whitened_ data_out
GFSK Modulator
Data_in
Modulator_out
rx_data_ out
start_tx
tx_data_in
tx_valid_in
Packet Framer / TK FSM
Start_tx
Tx_data_in
Tx_valid_in
Crc_in
Tx_ready_out
Irq_tx_done
Serial_data_out
Ctr_out
CRC-24 Gen
Data_in
enable
Crc_out

Block Diagram (Tx Path)

- The Host Controller Interface (HCI) provides payload data as Protocol Data Units (PDUs) to the baseband.
- The CRC-24 Generator computes a 24-bit checksum to ensure packet integrity during transmission.
- The Data Whitener randomizes the bit sequence using a 7-bit LFSR to avoid repetitive patterns and maintain DC balance.
- The Packet Framer appends preamble, access address, and synchronization fields to form a valid BLE packet.
- The GFSK Modulator and RF Transmitter convert the processed digital packet into a modulated signal for wireless transmission.

Receiver (RX) Path




Sync & AA Detector
bit_in
sync_found
Data de-whitener
data_in
enable
dewhitened_data
CRC-24 Checker
data_in
enable
Crc_ok
crc_check_en
Packet Parser / RX FSM
sync_found
bit_n
crc_ok
rx_data_put
rx_valid_out
irq_rx_done
ctrl_out
clk
rst_n
demod_in
GFSK Demodulator
Demod_in
Recovered_bit

Block Diagram (Rx Path)

- The RF Receiver captures the incoming BLE signal and converts it into a digital bitstream.
- The GFSK Demodulator restores the baseband data from the modulated waveform.
- The Access Address & Sync Detector identifies the start of a valid BLE packet using known synchronization patterns.
- The Data De-Whitener reverses the whitening process through a 7-bit Galois LFSR to recover the original data.
- The CRC-24 Checker and HCI Output verify packet accuracy and send valid payload data to the higher layers.

Top-Level Transmitter Integration

- The ble_tx_top module integrates all transmitter blocks into a single hierarchical design.
- It interfaces with the external host through handshake signals (tx_valid_in and tx_ready_out) for controlled data transfer.
- The Packet Framer coordinates CRC generation, data whitening, and serialization of the packet.
- All processed data is forwarded to the modulation chain for RF signal generation.
- An interrupt signal (irq_tx_done) is generated after successful packet transmission.

BLE Packet Framer

- The Packet Framer is implemented as a 7-state Finite State Machine that controls the complete transmission sequence.
- It generates the BLE packet structure consisting of Preamble, Access Address, Header, Payload, and CRC fields.
- The payload length is dynamically captured from the header to support variable packet sizes.
- It enables the CRC and Whitening blocks during the appropriate packet states.
- After payload transmission, it appends the computed 24-bit CRC and generates an interrupt upon completion.

CRC-24 Generator

- A 24-bit CRC generator is implemented using a Fibonacci LFSR architecture for error detection.
- The CRC computation is enabled during the Header and Payload transmission stages.
- Polynomial division is performed in real time as each bit is serialized.
- The final 24-bit remainder is provided to the Packet Framer for transmission.
- This ensures packet integrity and reliable error detection at the receiver.

Data Whitener (7-bit Galois LFSR)

- A 7-bit Galois LFSR is implemented to perform data whitening as required in BLE communication.
- The LFSR is seeded using the channel index at the start of each packet.
- Whitening is applied during the Header, Payload, and CRC transmission stages.
- Each outgoing bit is XORed with the LFSR output to randomize the bit pattern.
- This reduces long sequences of identical bits and improves transmission reliability.

Clock Scaling and Domain Management

- The baseband processing operates at a 20 MHz system clock.
- A Phase-Locked Loop (PLL) multiplies the clock frequency by 20 to generate a 400 MHz high-speed clock.
- The high-speed clock provides sufficient resolution for digital GFSK modulation.
- The modulator operates only after the PLL lock signal is asserted to ensure stable operation.
- This dual clock-domain architecture enables efficient baseband processing and high-speed modulation.

Digital GFSK Modulator (NCO Based)

- A fully digital GFSK modulator is implemented using a Numerically Controlled Oscillator (NCO).
- The phase accumulator generates frequency-shifted output based on the input data bit.
- Logic ‘1’ and Logic ‘0’ are represented using positive and negative frequency deviation from the carrier.
- Gaussian smoothing is approximated by gradually adjusting the frequency during bit transitions.
- The final output is a digitally modulated RF signal suitable for transmission.

GFSK Parameter Calculations and Validation

- The internal clock frequency is set to 400 MHz using a 20× PLL multiplier.
- The Samples Per Symbol (SPS) is calculated as 20, ensuring smooth modulation.
- NCO tuning word for 100 MHz carrier frequency is computed using an 8-bit phase accumulator.
- Frequency deviation of 10 MHz produces distinct frequency shifts for logic ‘1’ and ‘0’.
- Simulation waveforms verify the expected frequency and timing behavior.

Software Requirements

- **Iverilog -** Verilog Compiler.
- **Yosys** – Synthesizes Verilog RTL into gate-level netlist, bridging abstract design to physical components.
- **OpenROAD** – Automates floorplanning, placement, clock tree synthesis, and routing for full physical design.
- **Magic** – Performs Design Rule Check (DRC) and generates the final GDSII layout file for fabrication.
- **Netgen** – Compares final layout to synthesized netlist (LVS) to verify correct logic implementation.
- **Verilator** – Lints and analyzes Verilog code to catch syntax and quality issues early.
- **KLayout** – Graphical tool to view and inspect final GDSII layout, allowing visual confirmation and analysis.

Results & Discussions

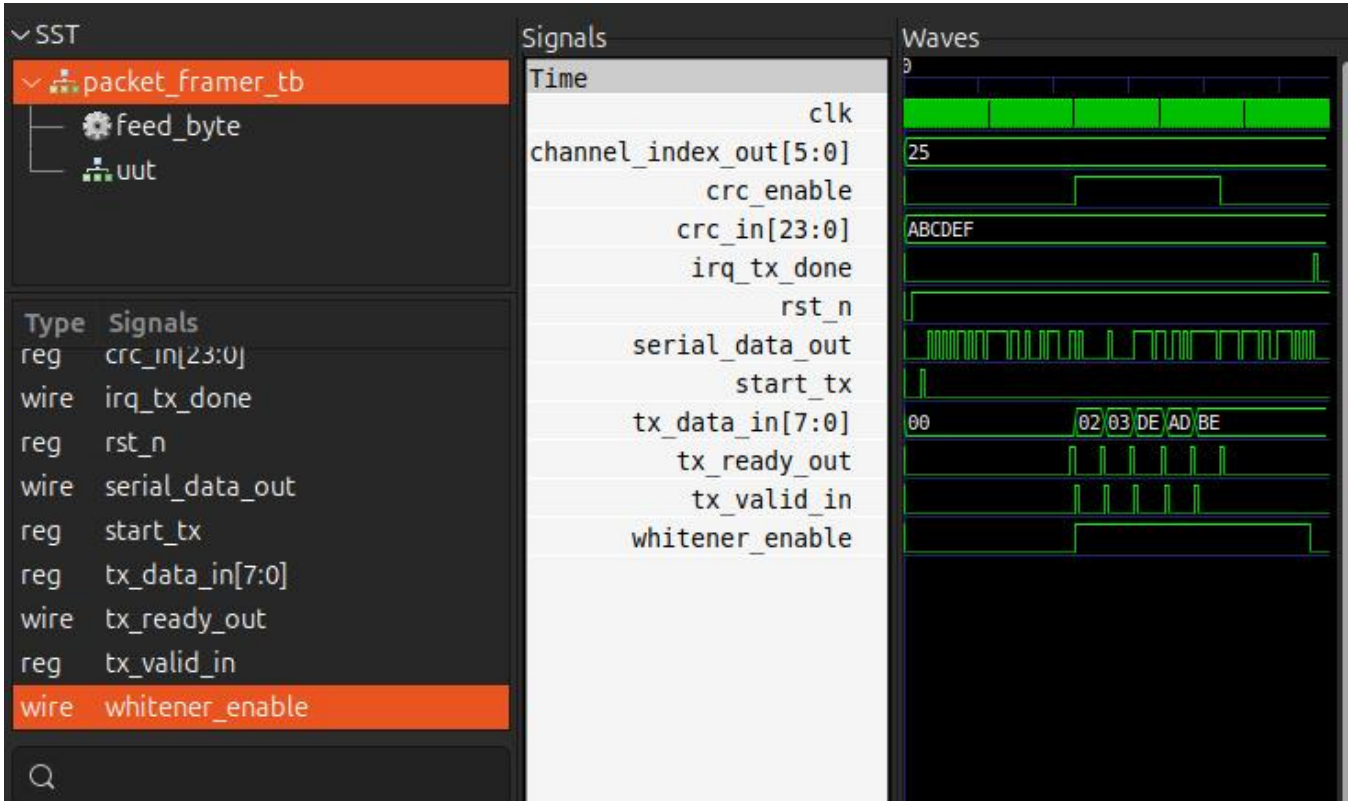


Fig: Successful compilation of the CRC block and it's testbench in I-verilog

Results & Discussions

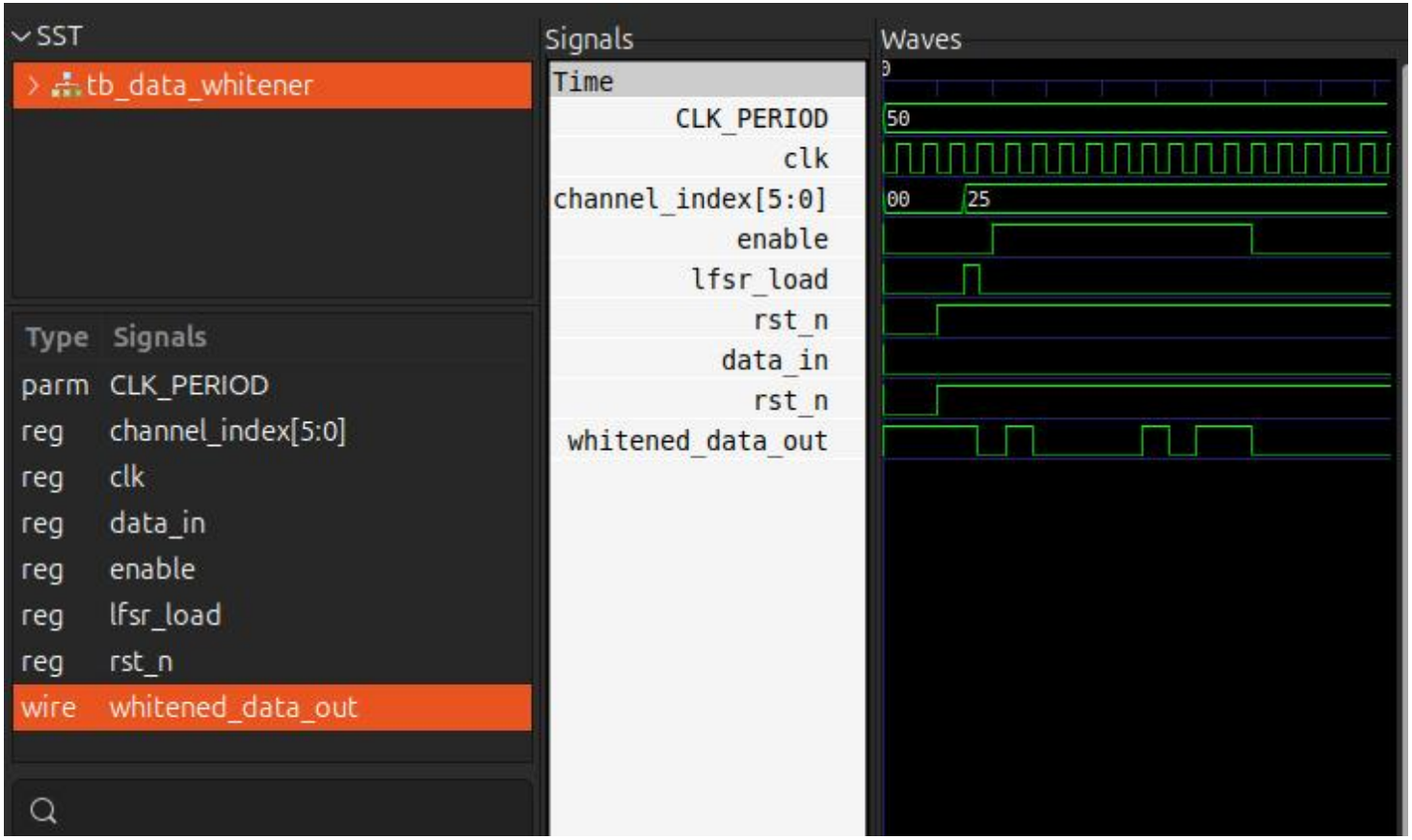


Fig: Waveforms of the CRC-24 block

Results & Discussions

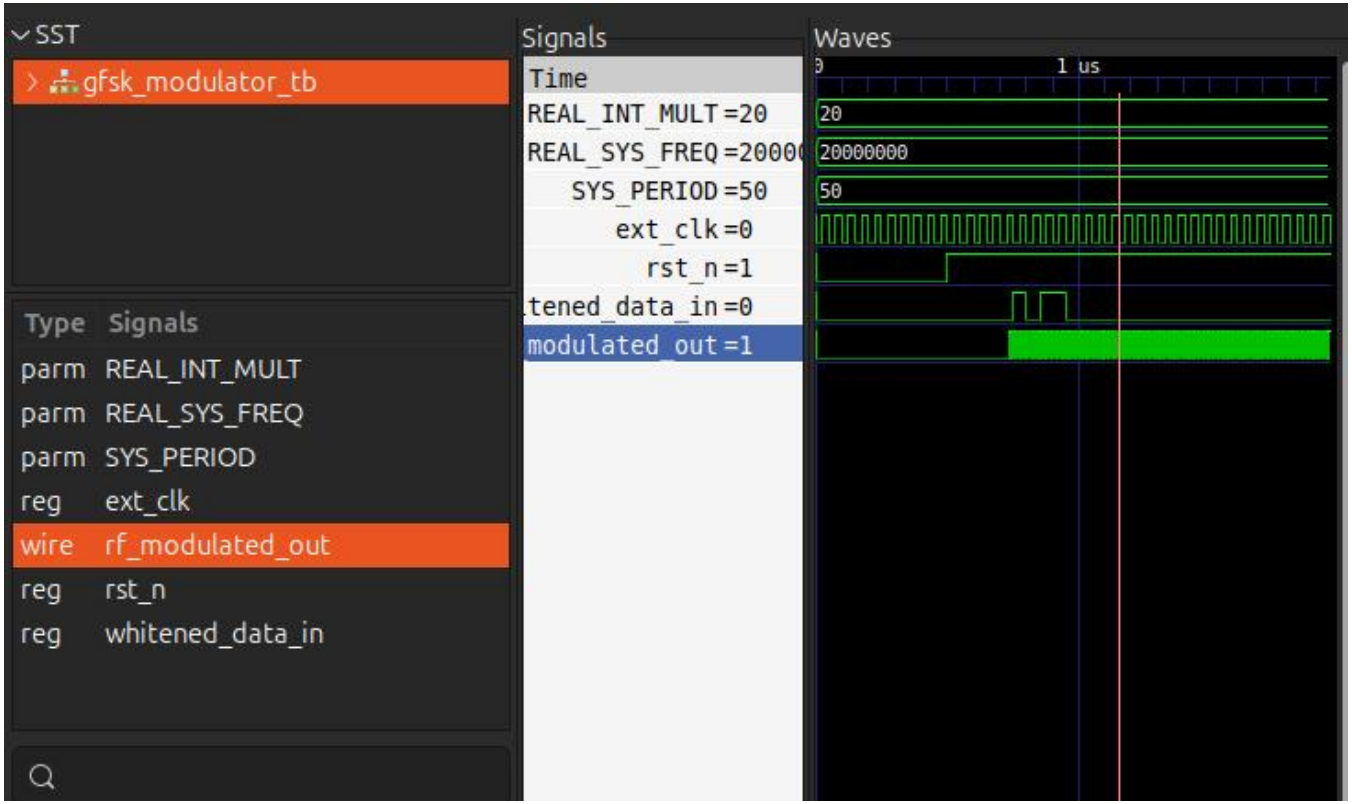


Fig: Successful compilation of the CRC block and it's testbench in I-verilog

Results & Discussions

- Data Whitener correctly loads channel index and generates pseudo-random sequence.
- whitened_data_out shows randomized bits, confirming proper XOR whitening operation.
- whitener_enable is active only during Header, Payload, and CRC fields.
- GFSK modulator generates frequency shift based on input bit (± Deviation).
- Output waveform shows different toggling density for ‘0’ and ‘1’, confirming modulation.
- Complete TX chain verified through simulation without functional errors.

Applications

- **Wearable Health Devices** – Enables low-power data transfer in fitness bands and smartwatches.
- **IoT Sensor Networks** – Supports efficient short-range communication between nodes.
- **Smart Home Systems** – Used for lighting, security, and automation control.
- **Industrial Automation** – Facilitates reliable wireless machine communication.
- **Automotive Systems** – Applied in keyless entry and sensor monitoring.
- **Consumer Electronics** – Integrated in wireless peripherals like earphones and controllers.
- **Healthcare Monitoring** – Enables continuous patient data transmission through BLE.

Conclusion

- A complete BLE digital transmitter was successfully designed and implemented using Verilog HDL.
- The system integrates Packet Framing, CRC-24 generation, Data Whitening, and NCO-based GFSK modulation.
- Dual clock-domain architecture enables efficient 20 MHz baseband processing and 400 MHz high-speed modulation.
- The design follows BLE packet structure and modulation requirements as per specification.
- Simulation results validate correct packet formation and frequency-shift keying behavior.
- The transmitter architecture is modular and scalable for future integration with the receiver path.

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Thank You!!!